

Introduction to Cloud Computing

Lecture 2 - Enabling technologies

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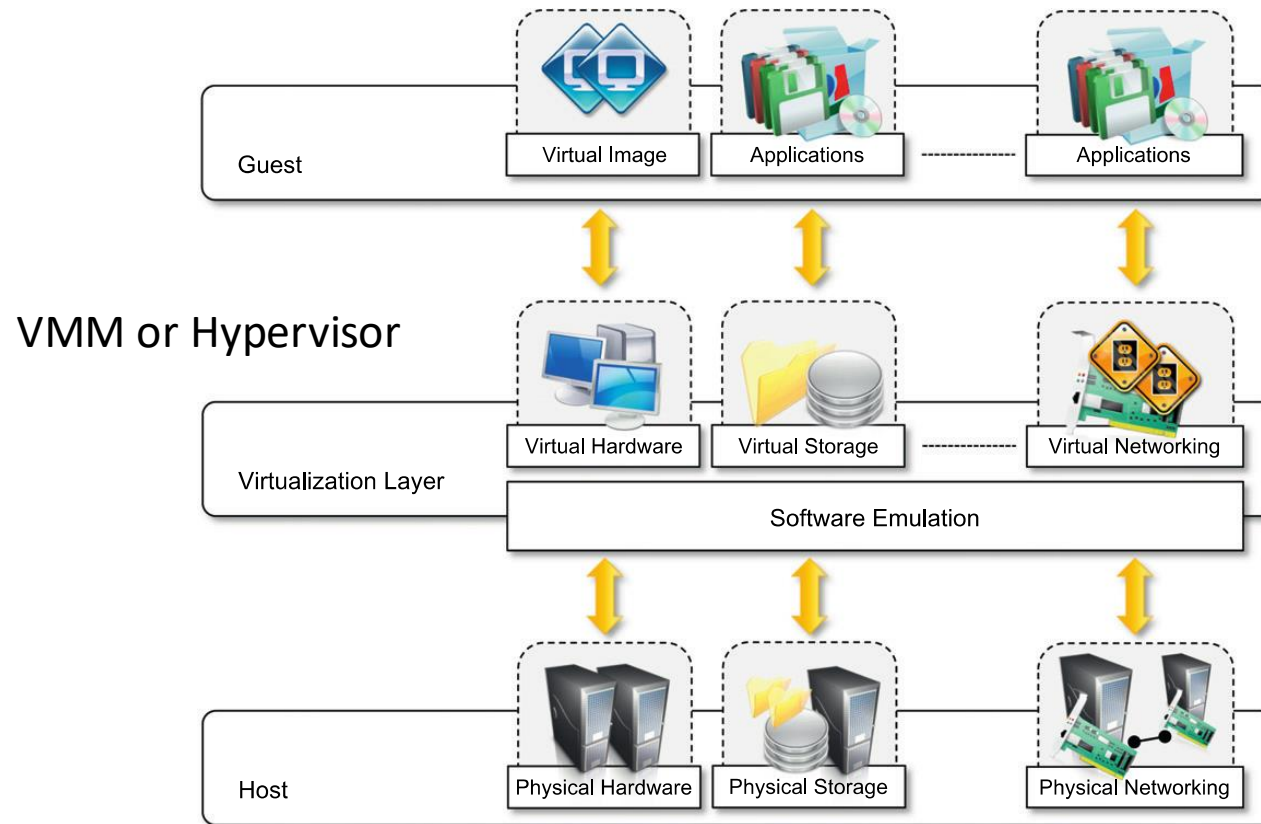
Agenda



- Main features of virtualization
- Basic concepts
- Virtual Machine Manager architecture and properties
- Virtualization techniques

Characteristics of virtualization environments

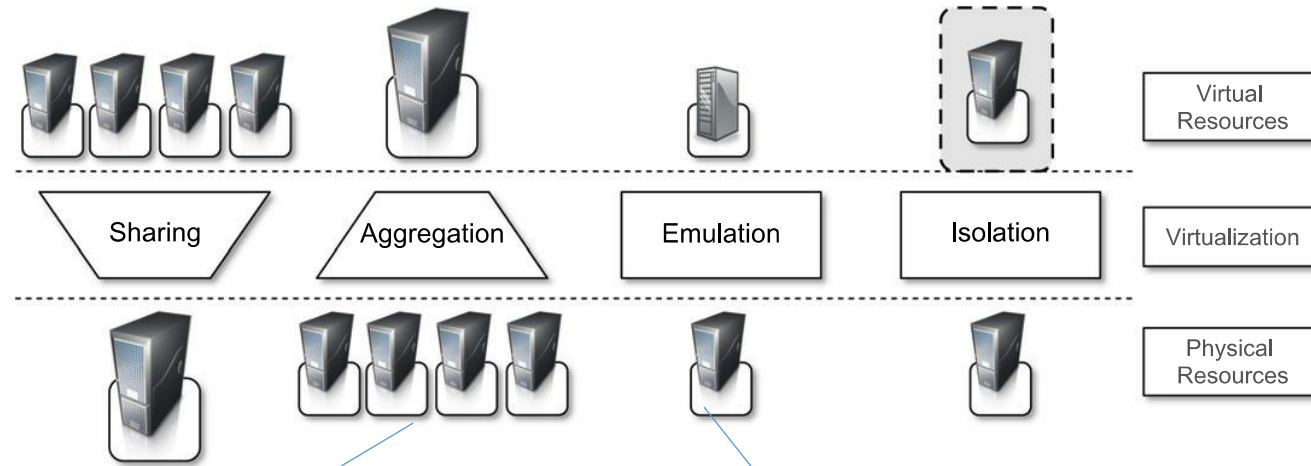
- Increased security
- Managed execution
- Portability



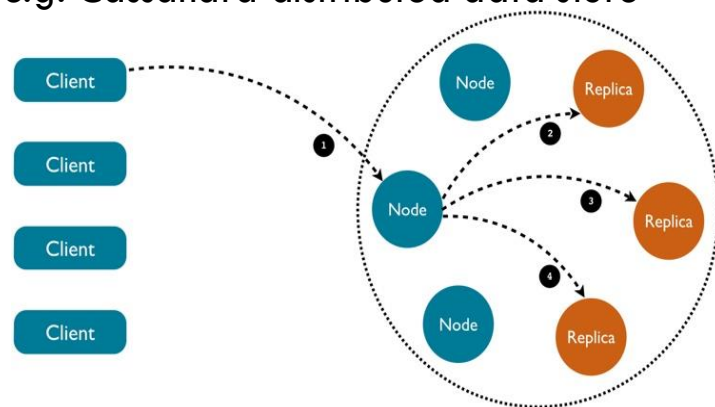
Increased security

- Why Virtualization increase security?
- The virtual machine manager control and filter the activity of the guest, thus preventing some harmful operations from being performed
 - Resources exposed by the host can then be hidden or simply protected from the guest
 - Sensitive information that is contained in the host can be naturally hidden without the need to install complex security policies
- Examples
 - In Hardware virtualization the file system exposed by the VM is completely separated from the file system of the host
 - The access to a memory location is granted only if it refers to the memory space allocated for the VM

Managed execution



e.g. Cassandra distributed data store



Portability

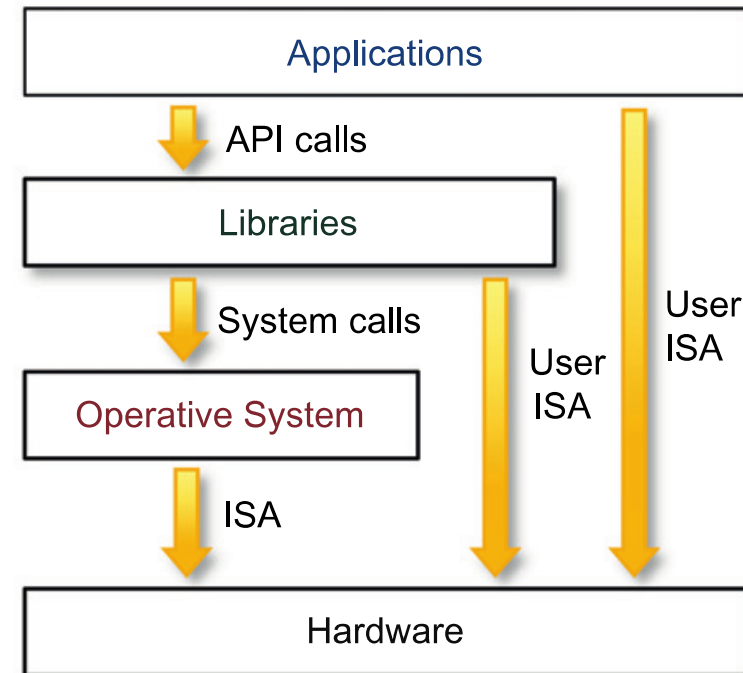
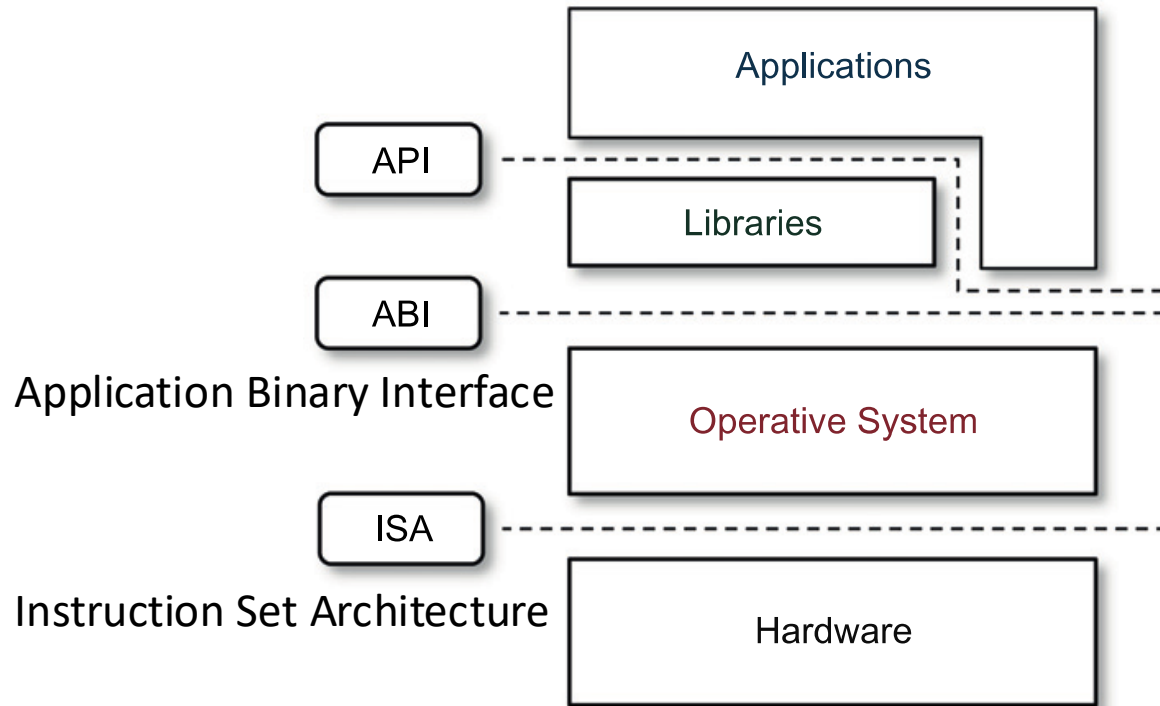
- Portability is the possibility to transfer and to use data and applications from a computing platform to another
- Why virtualization increase portability?

Portability

- A VM is booted from a VM image that is stored in a specific disk format
 - The disk format of a virtual machine image is the format of the underlying disk image, e.g.
 - aki, ari, ami, vdi, vhd, vhdx, vmdk, ...
 - You can move around VM keeping or converting their format.

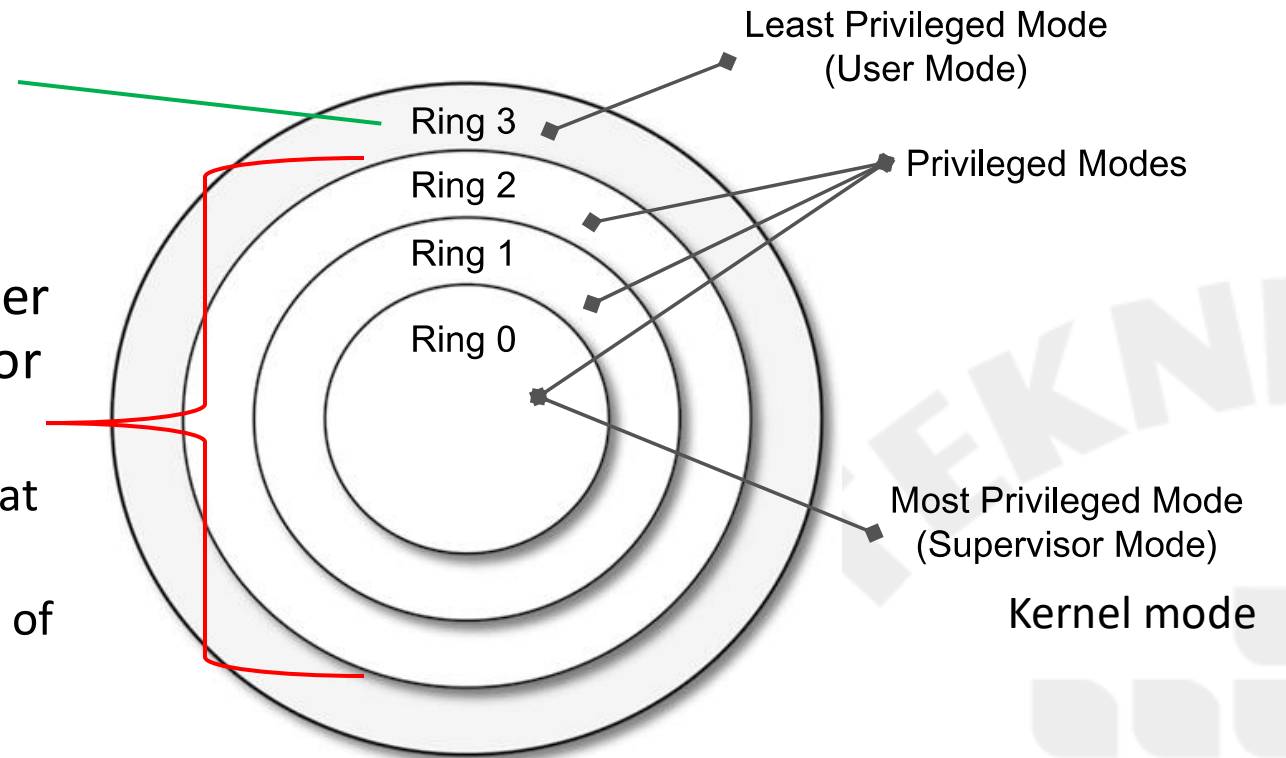
Machine reference model

- To understand how the different virtualization techniques work it is fundamental to recall the layered structure of a computer system



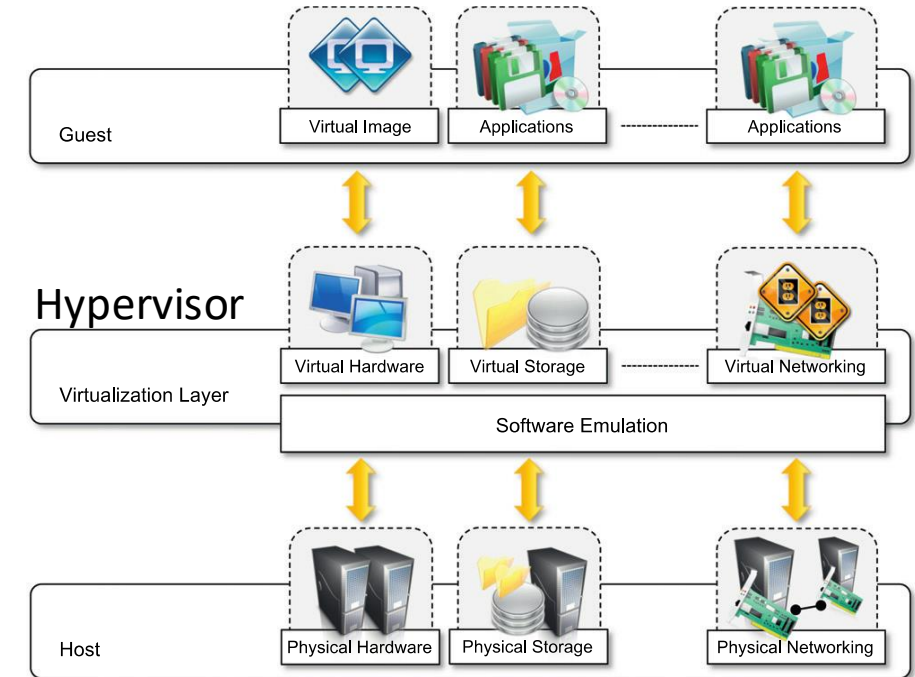
Privileged and Nonprivileged instructions

- **Nonprivileged** instructions - can be used without interfering with other tasks
 - floating, fixed-point, and arithmetic instructions.
- **Privileged** instructions - are executed under specific restrictions and are mostly used for sensitive operations
 - behavior-sensitive instructions are those that operate on the I/O
 - control-sensitive instructions alter the state of the CPU registers
- Ring 1 and 2 are not used in modern systems
 - We have only ring 0 and ring 3

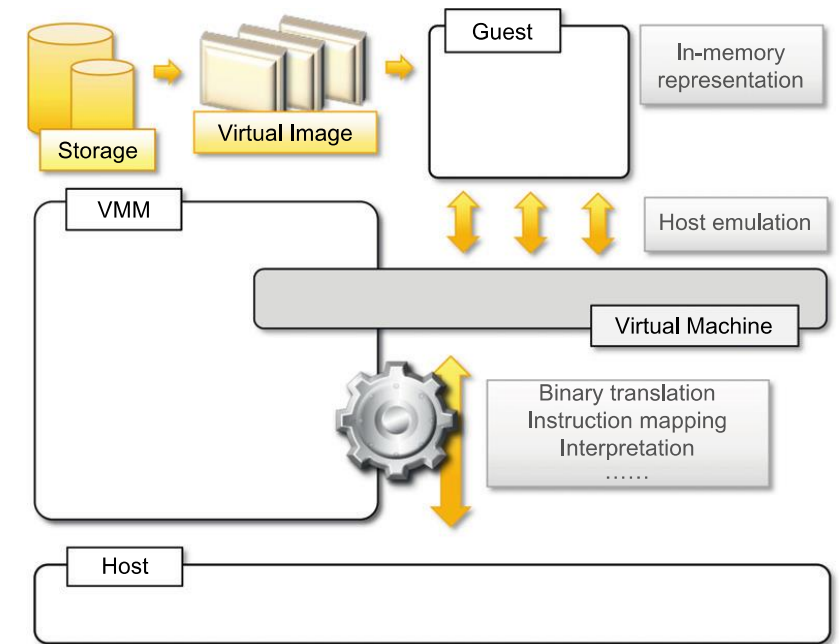
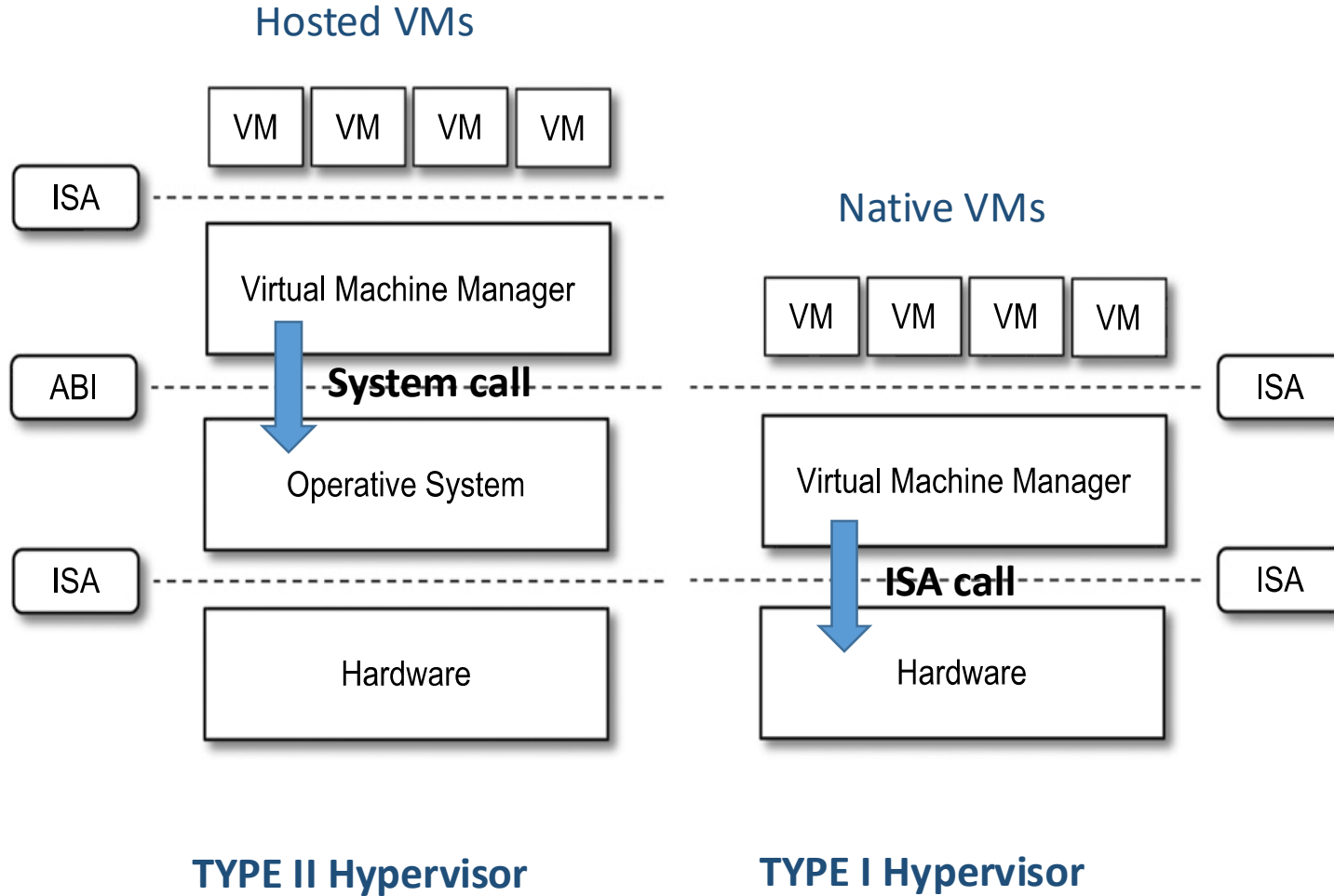


Hyper-visor

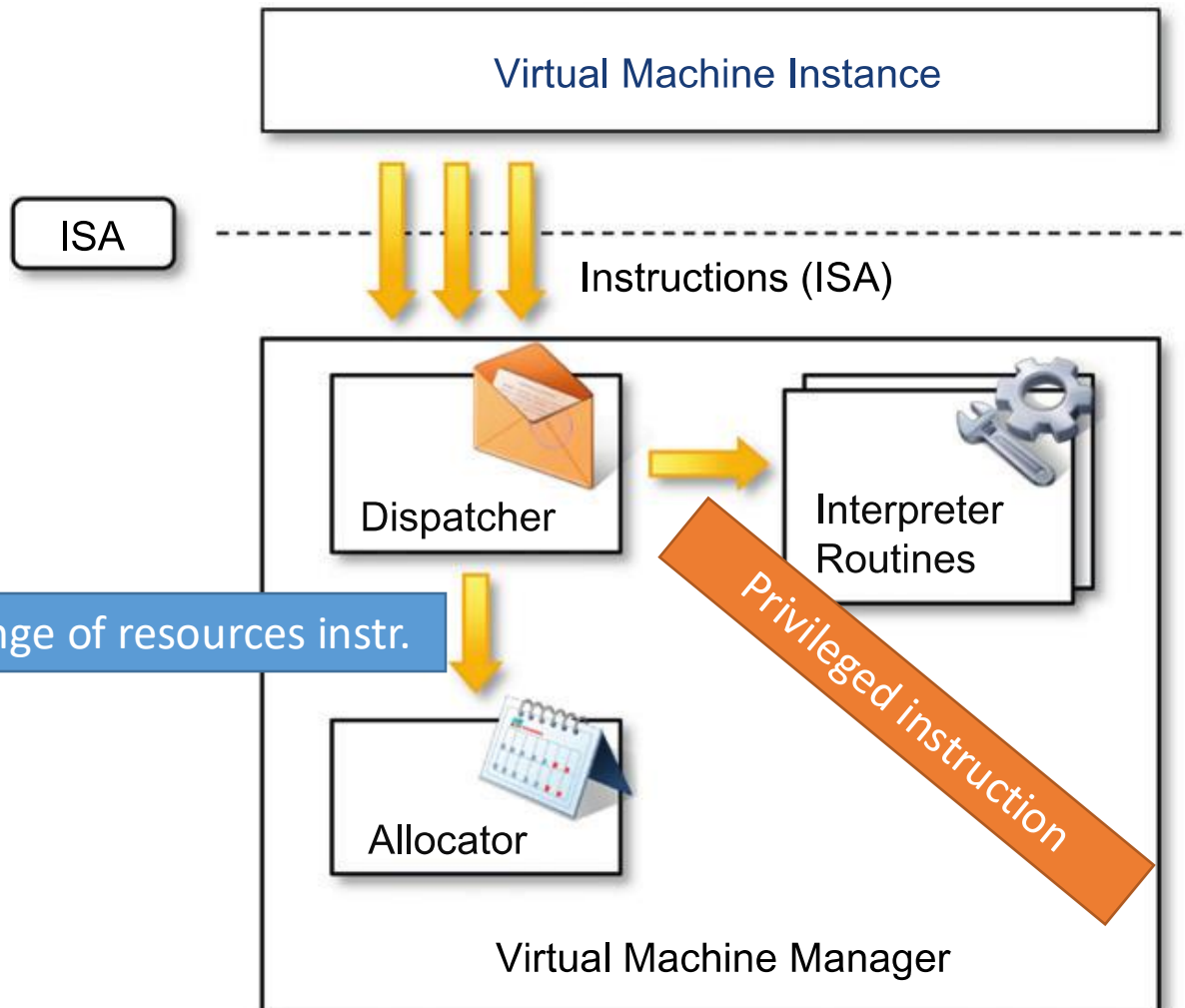
- The hyper-visor run on top of the supervisor mode, hence *hyper*
 - In practice, hypervisor run in supervisor mode
- Challenge: to fully emulate and manage the status of the CPU for guest operating systems all the sensitive instructions will be executed in privileged mode, which requires supervisor mode
 - Not true for the original ISA (17 sensitive instructions called in user mode) → no isolation among guest OSs
 - With Intel VT and AMD V such instructions have been redesigned as privileged



Hardware (or system) level virtualization



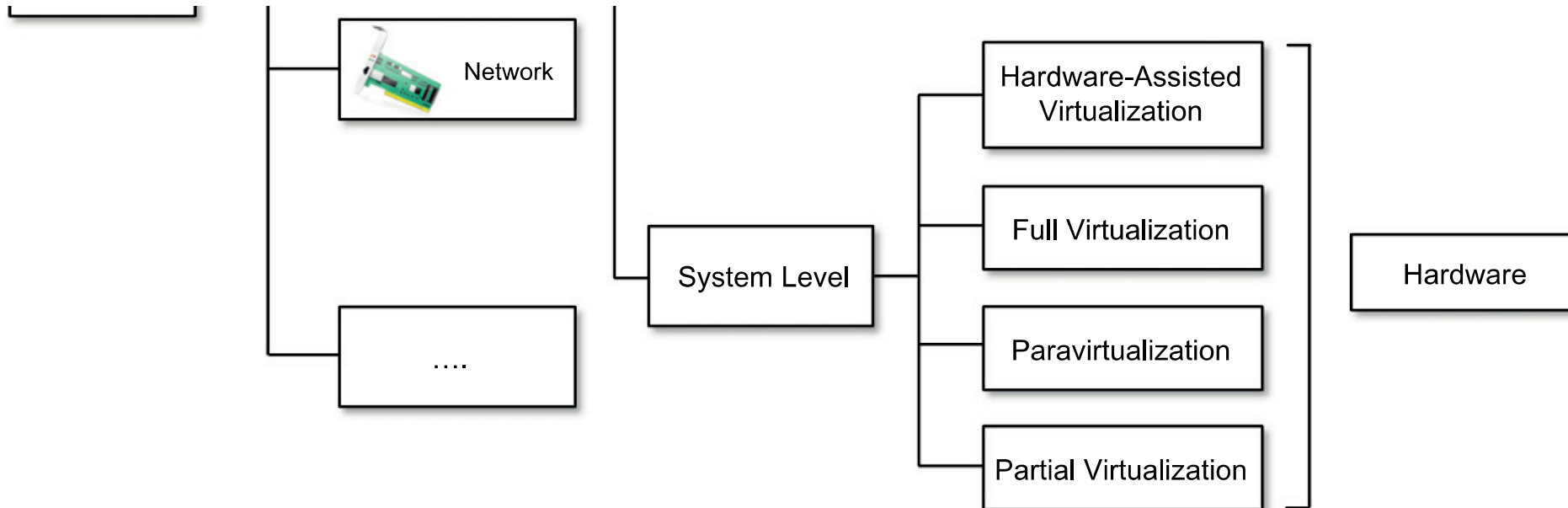
Virtual Machine Manager



Three properties must be satisfied

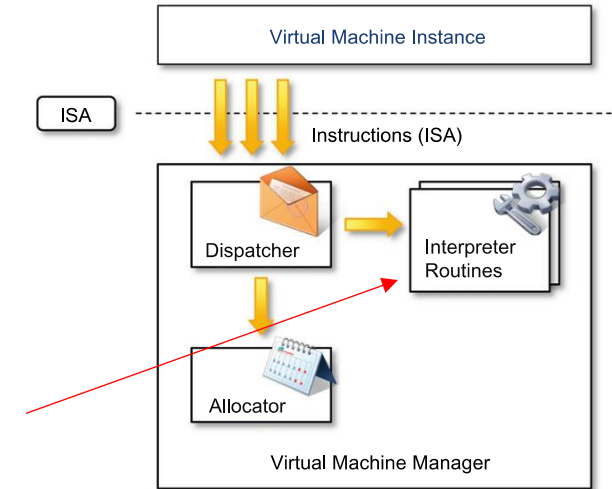
- **Equivalence.** A guest running under the control of a virtual machine manager should exhibit the same behavior as when it is executed directly on the physical host
- **Resource control.** The virtual machine manager should be in complete control of virtualized resources
- **Efficiency.** A statistically dominant fraction of the machine instructions should be executed without intervention from the virtual machine manager

Popek GJ, Goldberg RP. Formal requirements for virtualizable third generation architectures. Commun ACM 1974;17(7)



Full virtualization

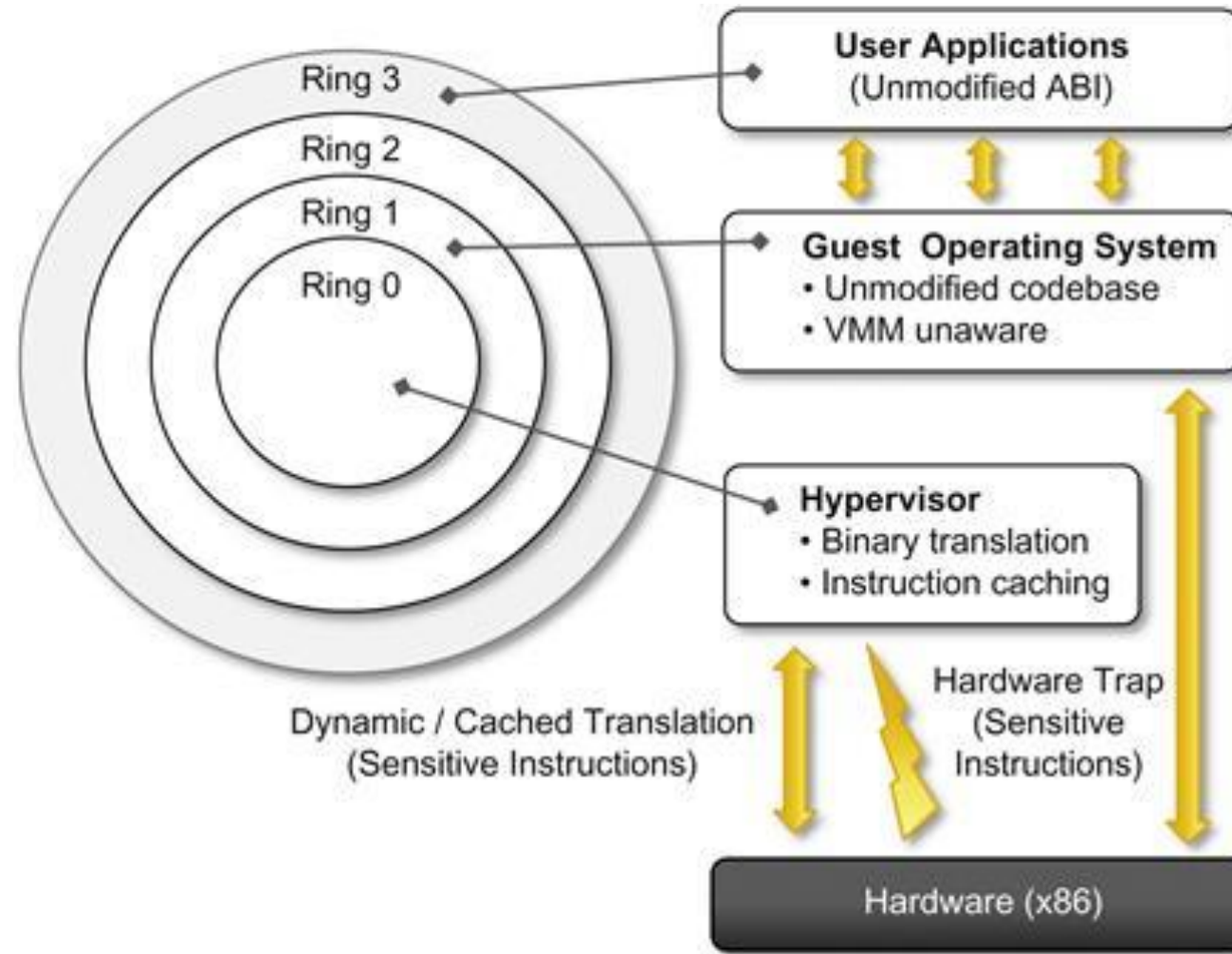
- The VMM scans the instruction stream
 - Noncritical instructions run on the hardware directly
 - Privileged, control- and behavior-sensitive instructions (critical) are identified and trapped into the VMM, which emulates the behavior of these instructions
 - Binary translation
- The guest OS is completely decoupled from the underlying hardware
 - the guest OS is unaware that it is being virtualized
- Binary translation is very costly especially for I/O operations



Binary Translation

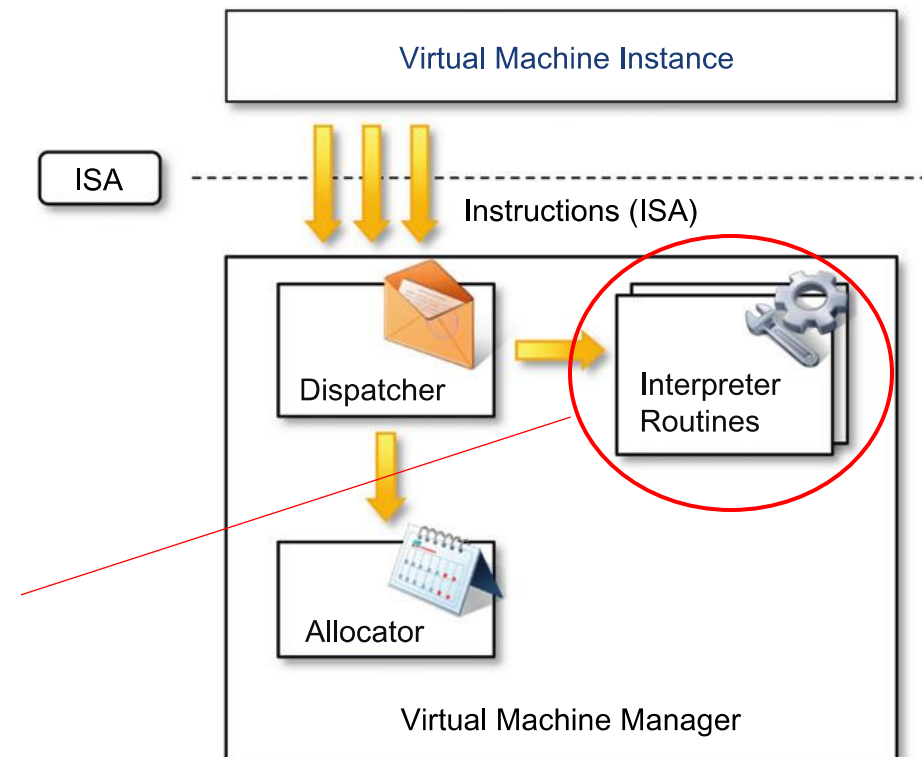
- Sequences of instructions are translated from a source instruction set to the target instruction set
- Static
 - aims to convert all of the code of an executable file into code that runs on the target architecture without having to run the code first
- Dynamic
 - looks at a short sequence of code, e.g. typically on the order of an instruction or short sequence of instructions, then translates it and caches the resulting sequence
 - Code is only translated as it is discovered and when possible, branch instructions are made to point to already translated and saved code
 - More overhead than static translation; re-execution of instruction and caching are leverage to make it more efficient

Full virtualization (cont'd)



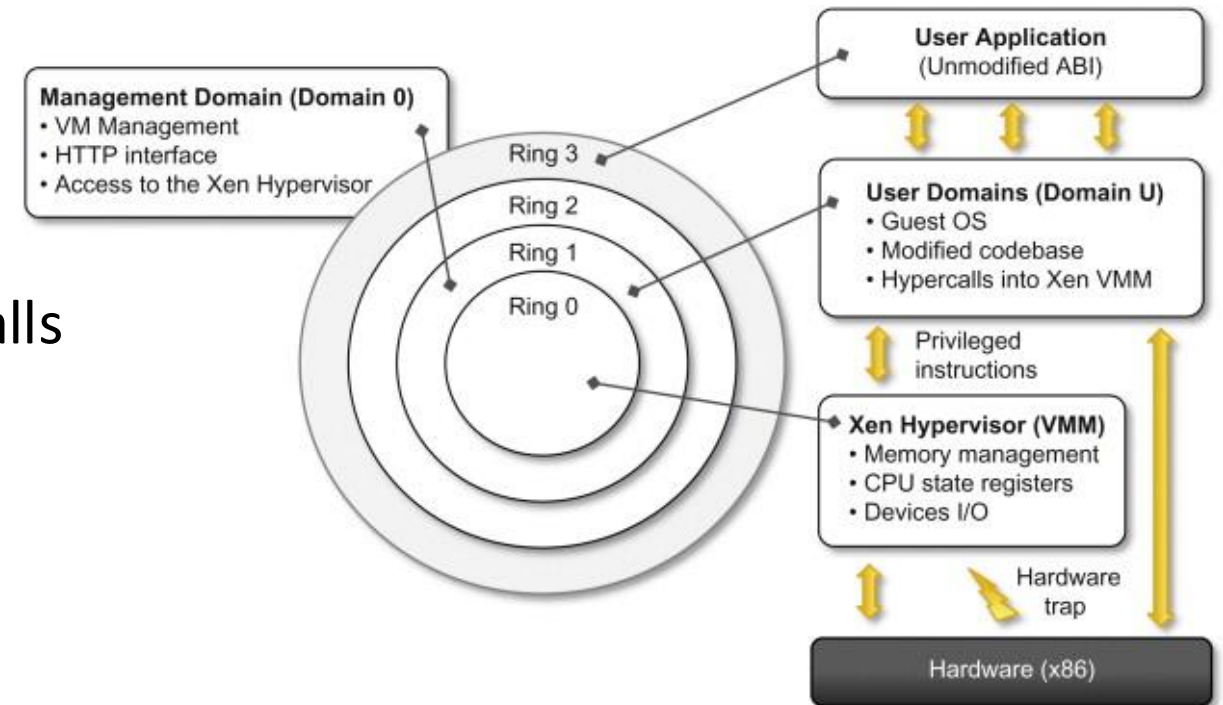
Hardware assisted virtualization

- Examples of hardware-assisted virtualization are
 - the extensions to the x86-64 bit architecture introduced with Intel VT
 - AMD V
- Reduce the performance penalties experienced by emulating x86 hardware with VMM
 - by design the x86 architecture did not meet the Equivalence requirements
 - 17 sensitive instructions called in user mode
 - early products were using binary translation to trap some sensitive instructions and provide an emulated version



Paravirtualization

- Not-transparent
 - A para-virtualized VM provides special APIs requiring substantial OS modifications in user domain
- Para-virtualized OS are assisted by an intelligent compiler to replace the nonvirtualizable OS instructions by hypercalls
- Para-virtualized OS prevents performance losses



To review the main concepts, answer the following questions



- Why binary translation is used in virtualization?
- How binary translation works?
- Why instruction caching is helpful in Full Virtualization?
- What are the main differences between HW assisted and not HW assisted virtualization?
- What are the differences between Paravirtualization and Full Virtualization?
- How Paravirtualization works

Assessment

- The concept presented in this lecture (Theme II) will be assessed in the final exam.

Assessment

When: Final Exam

Criteria: Analysis, Knowledge and understanding, Critical thinking, Language